Graph Theory – Lecture Notes

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Contents

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1 Introduction

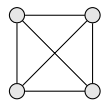
Definition 1.1. A *(simple) graph* G is a pair (V, E) where V is a non-empty set and $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$. We call the elements of V *vertices* of G and the elements of E *edges* of G. We also write $V(G)$ for V and $E(G)$ for E, respectively.

'Simple' means no loops and no double edges.

Remark 1.2. V is finite, unless otherwise stated.

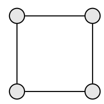
Example 1.3. We define a few common graphs. Let $n, m \in \mathbb{N}$.

- The *complete graph* K_n has n vertices and each pair of vertices is connected.



K_4

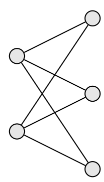
- The *cycle* C_n has n vertices, which are cyclically connected.



C_4

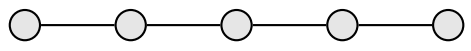
- Let A and B be disjoint sets with $|A| = n$ and $|B| = m$. The *complete bipartite graph* $K_{m,n}$ is defined by

$$V = A \dot{\cup} B \tag{1}$$
$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



$K_{2,3}$

- The *path* P_n has the vertices $1, \ldots, n$ and edges $\{a, a+1\}$ for $a \in \{1, \ldots, n-1\}$. (in books sometimes P_{n-1} has n vertices)



P_5

Definition 1.4. Let G be a graph, $u, v \in V(G)$

- If $\{u, v\} \in E(G)$, then u and v are **adjacent** and the edge $e = \{u, v\}$ is **incident** with both u and v.
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$ is the **neighbourhood** of v in G.
- $\deg_G(v) = |N_G(v)|$ is the **degree** of v.

Lemma 1.5 (Handshake Lemma)**.** *Let G be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \tag{3}$$

(in particular the sum of the degree is even).

Proof. $\sum\limits_{v \in V(G)} \deg_G(v) = \#\{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\} = 2|E(G)|$ $\square$

Definition 1.6. Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists an isomorphism ϕ between them, i.e. a bijection $\phi : V \to V'$ such that $\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G')$

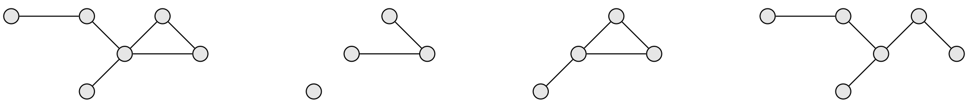
Definition 1.7. Let $G = (V, E)$ be a graph. A **subgraph** of G is a graph $H = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$ where $\binom{S}{K}$ is the set of all K-subsets of S.

H' is an **induced subgraph** of G if it is a subgraph and $E' = E \cap \binom{V'}{2}$

H'' is a **spanning subgraph** if it is a subgraph and $V' = V$

For $W \subseteq V$, write $G[W]$ for the subgraph induced by the vertices in W

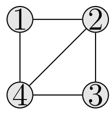
Example 1.8.



G H H' H''

Definition 1.9. Let v_i be vertices of a graph G

- A **walk** is a sequence of vertices v_i s.t. $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, ..., k-1\}$
- The **length** of a walk is k (number of vertices in the walk)
- The walk is **closed** if $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which $v_1, ..., v_k$ are pairwise distinct



$(1, 2, 3, 4, 2, 3)$ is a walk, but no trail
$(1, 2, 3, 4, 2)$ is a trail, no path
$(1, 2, 3, 4)$ is a path

Definition 1.10. Let G be a graph

- A graph G is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of G is an inclusion wise maximal connected subgraph

1.1 HamiltonianCycles

Definition 1.11. A **Hamiltonian cycle** in a graph G is a spanning subgraph that is a cycle (isomorphic to C_n for some $n \in \mathbb{N}$)
If G has a Hamiltonian cycle, G is called **Hamiltonian**

Theorem 1.12 (Dirac's Theorem)**.** *Let G be a graph of order $n \geq 3$ where the* ***order*** *of G is given by $|V(G)|$ and the* ***size*** *by $|E(G)|$. If every Vertex of G has degree at least $\frac{n}{2}$ then G is Hamiltonian*

Theorem 1.13 (Ore's theorem)**.** *Let G be a graph of order $n \geq 3$. If for every pair $u, v \in V, u \neq v, u$ and v non-adjacent, we have $\deg(u) + \deg(v) \geq n$ then G is Hamiltonian*

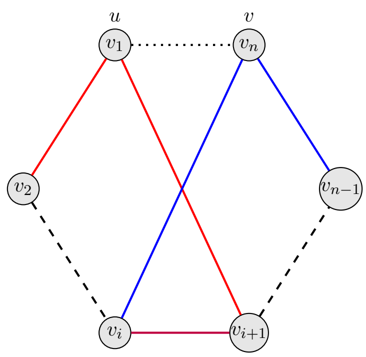
Remark 1.14.
Ore $\implies$ Dirac

Proof.

The Graph G is connected: suppose G is not connected. Take an arbitrary component of G and call it H. The order of H is $n-k$ if k vertices are not in $V(H)$. Now any $v \in V(H)$ has $\deg(v) \le n-k-1$ with equality if v is adjacent to every vertex in $V(H)$. Now take $u \notin V(H)$ so $\deg(u) \le k-1$. It follows that $u, v \in V(G), u \neq v$ and u, v are not adjacent so $\deg(u) + \deg(v) \ge n$ but $\deg(u) + \deg(v) \le k - 1 + n - k - 1 = n - 2 < n$ ↯

- G has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge $\implies$ G complete, hence Hamiltonian
- Assume G non complete $u, v \in V, u \neq v$, be non-adjacent. By induction the graph G' obtained from G by adding $e = \{u, v\}$ as an edge is Hamiltonian. Let C be a Hamiltonian cycle of G'. If C does not use e $\implies$ C is Hamiltonian cycle in G. Assume C uses e. Removing e from C yields a path from u to v

  For $i \in \{2, ..., n-1\}$
- If v_i is a neighbour of u, color $\{v_{i-1}, v_i\}$ red
- If v_i is a neighbour of v, color $\{v_i, v_{i+1}\}$ blue
- Since $\deg(u) + \deg(v) \ge n$ and the path $v_1, ..., v_n$ has $n-1$ edges, there exists an edge $\{v_i, v_{i+1}\}$ which is both red and blue. Then $v_1, ...v_i, v_n, v_{n-1}, ..., v_{i+1}$ is a Hamiltonian cycle in G.



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1.2 EulerianTrail

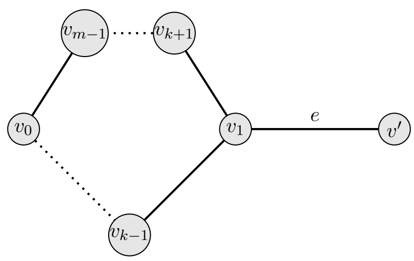
Definition 1.15. An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

Theorem 1.16 (Euler's theorem)**.** *Let G be a connected graph with at least 2 vertices. Then G has a closed Eulerian trail iff every vertex in G has even degree*

Proof.
" $\implies$ "
Let T be a closed Eulerian trail and fix an orientation on T. Then for every $v \in V(G)$, the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of T, so $\deg(v)$ is even.
" $\impliedby$ "
Let $T = (v_0, e_1, v_1, ..., e_m, v_m)$ be a longest trail. We will prove (i) $v_0 = v_m$ and (ii) $\{e_i : i \in \{1, ..., m\}\} = E(G)$. (i)+(ii) $\iff$ T is closed Eulerian trail.

(i) If $v_0 \neq v_m$, then v_0 is incident to an odd number of edges in T. As $\deg(v_0)$ is even, there is $e \in E(G)$ incident to v_0 and not on T. We could thus append e on T $\lightning$ because T is longest trail.

(ii) Claim: $V(T) = V$. Assume otherwise there exists a vertex $v' \notin V(T)$ and $e = \{v_k, v'\} \in E(G)$ for some $k \in \{0, ..., m\}$ (by connectedness). Then $(v', e, v_k, e_{k+1}, v_{k+1}, ..., v_{m-1}, e_m, v_0, e_1, ..., v_{k-1}, e_k)$ is a longer trail hence $V(T) = V$



Claim: $E(T) = E$. Assume otherwise. Consider $e \in E \setminus E(1)$ and write $e = \{v_k, v_l\}$. Then $(v_k, e_{k+1}, v_{k+1}, ..., v_{m-1}, e_m, v_0, e_1, v_1, ..., e_k, v_k, e, v_l)$ is a longer trail $\lightning$

□

Theorem 1.17. *Let G be a connected graph with at least two vertices. Then G has an Eulerian trail iff there are precisely 0 or two vetices of odd degree*

Proof.
By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

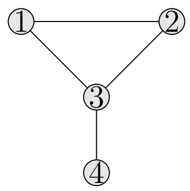
Suppose now that G has exactly two vertices x and y of odd degree. We consider the following graph G': Set $V(G') = V(G) \cup \{z\}$, where $z \notin V(G)$. Furthermore set $E(G') = E(G) \cup \{\{x, z\}, \{y, z\}\}$. Clearly, every vertex has even degree in G', so there exists a closed Eulerian trail. Since z is only adjacent to x and y, these are the neighbours of z on the trail. Deleting the edges $\{x, z\}$ and $\{y, x\}$ from the trail yields an Eulerian trail from x to y in G. Conversely, if there is a non-closed Eulerian trail with endpoints x and y in G, then x and y are non-adjacent. Adding the edge $\{x, y\}$ creates a graph with a closed Eulerian trail, so all vertices have even degree by Euler's theorem. Thus x and y are the only verices in G with odd degree. □

1.3 Datastructures

Assume that $G = (V, E)$ is a graph with $v = \{1, ...n\}$

Definition 1.18. The **adjacency matrix** of G is the matrix $A = (a_{ij})$ with
$$a_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & else \end{cases}$$
The **adjacency list** of G stores a label for each vertex, and a list of (pointers to) its neighbours.

Example 1.19.



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- v_1: v_2, v_4

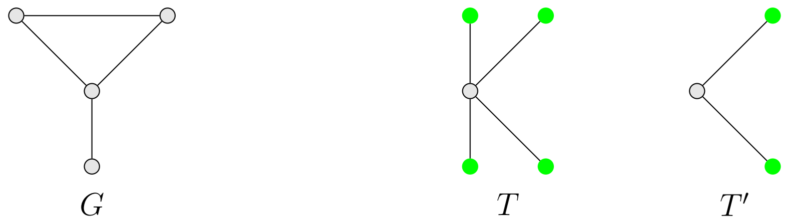
- v_2: v_1, v_3, v_4
- v_3: v_2
- v_4: v_1, v_2

Remark 1.20.
Memory requirement:
Adjacency matrix: $\theta(n^2)$
Adjacency list: $\theta(n+m)$, where m is the number of edges

2 Trees

Definition 2.1. A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to C_n for some $n \geq 3$).
A **forest** is a graph that does not contain a cycle.
A **leaf** is a vertex of a tree with degree 1.

Example 2.2.



- G is not a tree nor a forest
- T, T' are trees and forests
- $T \cup T'$ is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

Lemma 2.3. *Every tree T with $n \geq 2$ vertices has at least one leaf.*

Proof. We iteratively construct a path $v_1, ..., v_n$ as follows:
Start with any edge $\{v_1, v_2\} \in E(T)$. Assume that we have constructed $v_1, ..., v_k$. To construct v_{k+1}, we may assume that v_k is not a leaf in T (or we are done). Otherwise v_k has a neighbour $v_{k+1} \neq v_{k-1}$. $v_{k+1} \neq v_i$ because T is a tree and otherwise we would get a cycle $(v_i, ..., v_k, v_{k+1} = v_k)$. Add v_{k+1} to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf. □

Theorem 2.4. *Let G be a graph with n vertices. The following are equal:*

(i) *G is a tree*

(ii) *G is connected with $n - 1$ edges*

(iii) *G is connected and removing an edge disconnects it*

(iv) *G has exactly one path between each pair of distinct vertices*

(v) G has no cycle, but adding an edge creates a cycle

Proof. "1. $\implies$ 2."
Induction on n:
$n = 1$ is trivial
Let $n \geq 2$. By Lemma (2.2), G has a leaf v. Removing v and the unique edge incident to v yields a tree G'. By induction G' has $n-2$ edges so G has $n-1$ edges.

"2. $\implies$ 3."
Suppose there is an edge $e \in E(G)$ and the resulting graph by removing e is still connected. Keep removing as long as the resulting graph is still connected. Let G' be the resulting graph. Observe that G' is a spanning subgraph of G. G' does not contain a cycle as otherwise we could remove one edge from the cycle. By "1. $\implies$ 2.", applied to G', we know that G' has $n-1$ edges $\lightning$ because G had $n-1$ edges and we removed at least one.

"3. $\implies$ 4."
Suppose that there is a pair with $u, v \in V(G)$ s.t. there are two distinct paths u_i, v_i between them. Let i_0 be the maximum index s.t. $u_i = v_i : \forall i \in \{1, ..., i_0\}$. The edge $\{v_{i_0}, v_{i_0+1}\}$ does not lie on the path u_i. The graph obtained by removing $\{v_{i_0}, v_{i_0+1}\}$ is connected as for any walk using this edge we can use u_i to go to u to reach v_{i_0} or v to reach v_{i_0+1}.

"4. $\implies$ 5.
If G had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).
Adding an edge creates a cycle: Let $u, v \in V(G)$ non adjacent. There is one path between the two. Now adding an edge between u, v creates a second path $\lightning$

"5. $\implies$ 1.
We only need to show that G is connected. Suppose u, v are not connected by a walk. Creating an edge between u, v does not result in a cycle because for being isomorphic to C_n one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting u, v is the newly added edge $\lightning$ $\square$

Corollary 2.5. *Every connected graph G has a spanning tree.*

Proof.
Consider graphs on n vertices, now do induction on the number of edges m. As G

is connected, $m \geq n-1$. If $m = n-1 \implies G$ is a tree. If $m > n-1 \implies G$ is not a tree. By (2.3) we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of G. $\square$

Theorem 2.6 (Matrix-tree theorem)**.** *Let G be a graph with vertices $\{1, ...n\}$, edges $\{e_1, ..., e_m\}$.*
Let $Q = (q_{ij})$ with $q_{ij} = \begin{cases} -1 & \{i,j\} \in E \\ 0 & i \neq j : \{i,j\} \notin E \\ \deg(i) & i = j \end{cases}$ Let Q_{ij} be the matrix obtained from Q by removing the i-th row and the j-th column. Then the number of labelled spanning trees of G is $|\det(Q_{ij})|$, for any choice of i, j.

Corollary 2.7 (Cayley's formula)**.** *The number of labelled spanning trees of K_n is n^{n-2} if $n \geq 2$.*

Proof. Define Q as in (2.5).

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \cdots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \cdots & 1 & n-1 \end{pmatrix} \to \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -n & n & 0 & \cdots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \cdots & 0 & n \end{pmatrix} \to \hat{Q}_{11} =$$
$$\begin{pmatrix} 1 & \star & \cdots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \cdots & 0 & n \end{pmatrix}$$
$$\implies |\det(Q_{11})| = |\det\left(\hat{Q}_{11}\right)| = n^{n-1-1} = n^{n-2}$$ $\square$

3 PlanarGraphs

Definition 3.1. A **planar drawing** of a graph G consists of injective continuous functions

$$\phi_v : V(G) \to \mathbb{R}^2 \qquad\qquad \phi_e : [0,1] \to \mathbb{R}^2$$

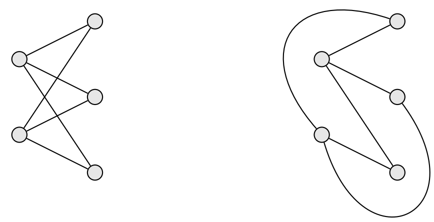
s.t.:

- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$ for all $\{u,v\} \in E(G)$
- The sets $\phi_e((0,1))$ for each $e \in E(G)$ are pairwise disjoint and disjoint from $\mathrm{Im}(\phi_v)$ (i.e. no crossings)

A graph G is called **planar** if it has a planar drawing.
A **plane graph** is a planar graph, together with a fixed planar drawing.

Example 3.2.

- Trees
- K_4
- $K_{2,3}$



Definition 3.3. A **face** of a plane graph G is a maximal connected part of $\mathbb{R}^2 \backslash (\mathrm{Im}(\phi_v(u)) \cup \mathrm{Im}(\phi_e))$. Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of G incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

Theorem 3.4 (Euler's formula). *Let $G = (V, E)$ be a connected planar graph. Then the number of faces $|F|$ of G is $|F| = |E| - |V| + 2$.*

Proof. Induction on $|E|$.
Since G is connected, we have $|E| \geq |V| - 1$ so we start the induction at $|E| = |V| - 1 \implies G$ is a tree. Since a tree does not contain a cycle, we have $|F| = |E| - |V| + 2$. Now let $|E| > |V| - 1$. Then G is not a tree, so G contains a cycle C. Choose an edge e on C and remove it to obtain a plane graph G'. e separates two faces in G, which fuse to a single face in G'. By induction, the number of faces $|F'|$ in G' is $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$. □

Lemma 3.5. *For every plane graph $G = (V, E)$ with $|V| \geq 3$, we have $|E| \leq 3|V| - 6$.*

Proof. Wlog G is connected. Consider a planar drawing of G and let F be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula (3.4), we obtain $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$. □

Remark 3.6. This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

Example 3.7.

- K_5 is not planar (10 edges) due to (3.4)
- $K_{3,3}$ is not planar (without proof)

Definition 3.8. A **subdivision** of a graph G is a graph obtained from G by repeatedly subdividing some of its edges.

Theorem 3.9 (Kuratowski's theorem). *A graph G is planar iff G contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.*

Remark 3.10. A subdivision of a graph is planar iff the graph is planar

Definition 3.11. Let G be a graph. The graph obtained from G by contracting $e = \{u, v\} \in E(G)$ denoted G/e, is the graph obtained from G by deleting u, v, inserting a new vertex w and making w adjacent to all vertices in $N_G(u) \cup N_G(v)$. A graph G' is a **minor** of G if G' can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

Remark 3.12. If G' is a minor of G then G' planar $\Longleftarrow$ G is planar.

Theorem 3.13 (Wagner's theorem)**.** *A graph G is planar iff $K_{3,3}$ and K_5 are not a minor of G.*

4 Flows,Cuts,Connectivity

Definition 4.1. A **directed graph** is a pair (V, E) with $E \subseteq (V \times V) \backslash \{(x, x) : x \in V\}$. Write $\delta_G^+(v)$ for edges starting in v and $\delta_G^-(v)$ for edges ending in v.

Definition 4.2. A **flow network** is a tuple (G, s, t, c) where G is a directed graph, $s, t \in V(G), s \neq t$ and $c : E \to \mathbb{R}_0^+$.

- s is called a **source**
- t is called a **target**
- $\forall e \in E(G) : c(e)$ is the **capacity of e**

A **flow** in a flow network is a function $f : E \to \mathbb{R}_0^+$ s.t.

- $f(e) \leq c(e)$ for all $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum\limits_{e \in \delta^-(v)} f(e) = \sum\limits_{e \in \delta^+(v)} f(e)$ (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$ (i.e. nothing flows into a source or out of a target)

The **value** of a flow f is $\text{val}(f) = \sum\limits_{e \in \delta^-(v)} f(e) = \sum\limits_{e \in \delta^+(v)} f(e)$

Remark 4.3. For $v \in V$, write $f(\delta^+(v)) = \sum\limits_{e \in \delta^+(v)} f(e)$; $f(\delta^-(v)) = \sum\limits_{e \in \delta^+(v)} f(e)$

Definition 4.4. A **cut** in a graph/directed graph G is a partition of $V(G)$ into two non-empty parts A and B (i.e. $V(G) = A \dot{\cup} B$ and $A, B \neq \emptyset$). For $s, t \in V(G), s \neq t \implies (A, B)$ is a (s, t)-cut if $s \in A, t \in B$.
If (S, T) is a (s, t)-cut, write $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$ and $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$.
The **capacity** of the cut is $c(S, T) = \sum\limits_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$.

Theorem 4.5 (max flow-min cut theorem)**.** *Let (G, s, t, c) be a flow network. Then* $\max\{\text{val}(f) : f \textit{ flow}\} = \min\{c(S, T) : (S, T) \textit{ is a } (s, t)\textit{-cut}\}$.

Lemma 4.6. *Let f be a flow, (S,T) be a (s,t)-cut. Then* $\mathrm{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$.

Proof.
Claim: $\forall A \subset V : f(\delta^+(A)) - f(\delta^-(A)) = \sum\limits_{v \in A} f(\delta^+(v)) - f(\delta^-(v)) = (*)$
The sum $f(\delta^+(A)) - f(\delta^-(A))$ does not account for edges $e \in E(G)$ with start and end in A. On the right side, $f(e)$ is counted twice, once with positiv and once with negativ sign, so the sum vanishes.
Then $\mathrm{val}(f) = f(\delta^-(t)) \stackrel{(1)}{=} \sum\limits_{v \in T} f(\delta^-(v)) - f(\delta^+(v)) \stackrel{(*)}{=} f(\delta^-(T)) - f(\delta^+(T)) = f(\delta^+(S)) - f(\delta^-(S))$
(1) holds because $f(\delta^+(t)) = 0$ (t is a sink) and $\forall v \notin \{s,t\} : f(\delta^-(v)) - f(\delta^+(v)) = 0$ (by definition). □

Corollary 4.7. *Let f be a flow, (S,T) an (s,t)-cut.*
$\mathrm{val}(f) \leq c(\delta^+(S))$
Thus $\max\{\mathrm{val}(f) : f \textit{ flow}\} \leq \min\{c(S,T) : (S,T)\ (s,t)\textit{-cut}\}$.

4.5.
max flow $\leq$ min cut due to Corollary (4.7)
Let f maximum flow (i.e. a flow for which $\mathrm{val}(f)$ is maximal). Let $c' : E(G) \to \mathbb{R}_0^+$ where $c'(e) = c(e) - f(e)$. Let $S \subset V(G)$ be defined as the set of reachable vertices[1] according to the following rules:

- $s \in S$
- $x \in V(G)$ is in S (i.e. reachable) if there is a walk (in the underlying directed graph) from s to x traversing $e_1, ..., e_k$ s.t.
  (i) $c'(e_i) > 0$ ($\iff f(e_i) < c(e_i)$) if e_i is traversed towards x.
  (ii) $c'(e_i) < c(e_i)$ ($\iff f(e_i) > 0$) if e_i is traversed towards s.

If $S = V(G)$, we can improve f as follows:
Let $e_1, ...e_k$ be a walk from s that makes t reachable, and e_i is traversed in orientation $\sigma_i \in \{\pm 1\}$, then $\tilde{f}(e_i) = f(e_i) + \sigma_i \varepsilon$, for $i \in \{1, ..., k\}$ and $\tilde{f}(e_i) = f(e_i)$ otherwise. This defines a flow $\tilde{f}$ with $\mathrm{val}(\tilde{f}) > \mathrm{val}(f)$ for $\varepsilon > 0$ sufficently small $\lightning$
Thus $t \notin S$. Set $T = V \setminus S \neq \emptyset$

- $\forall e \in \delta^+(S) \quad c'(e) = 0$, i.e. $f(e) = c(e)$
  $e = \{x, y\}$ with $x \in S, y \in T$. If $c'(e) \neq 0 \implies e$ satisfies $(i) \implies y$ reachable $\lightning$

[1] These are the vertices where the flow "is not maximal yet" but has a walk to s.

- $\forall e \in \delta^-(S) \quad c'(e) = c(e) \iff f(e) = 0$

- $e = \{y, x\}$ with $x \in S, y \in T$. If $c'(e) \neq c(e) \implies e$ satisfies $(ii) \implies y$ reachable $\lightning$

- $c(S,T) = \sum_{e \in \delta^+(S)} c(e) = \sum_{e \in \delta^+(S)} f(e) - \sum_{e \in \delta^-(S)} f(e) = f(\delta^+(S)) - f(\delta^-(S)) \overset{(4.6)}{=}$ $\text{val}(f) \implies \text{val}(f) \geq c(S,T) \geq \min(s,t)$-cut

□

4.1 Connectivity

Definition 4.8. A graph G is **k-vertex connected** ($k \in \mathbb{N}$) if it has at least $k+1$ vertices and whenever we remove up to $k-1$ vertices, the resulting graph is connected.
A graph G is k-**edge connected** if whenever we remove up to $k-1$ edges, the resulting graph is connected.
The **vertex connectivity (edge connectivity)** is the largest $k \in \mathbb{N}$ s.t. G is k-vertex/edge connected.

Remark 4.9. G graph, k vertex connectivity,λ edge connectivity,δ minimal degree of G, then $k \overset{(i)}{\leq} \lambda \overset{(ii)}{\leq} \delta$

Proof. (i) If we have a set of λ edges whose removel makes the res. graph disconnected, then deleting one endpoint for eah edge also disconnects the graph.

(ii) Delete all edges incient to a vertex of minimal degree,this disconnects the graph.

□

Lemma 4.10. *G directed graph, $s, t \in V, s \neq t$. Then the maximal number of edge-disjoint (s,t)-paths (path which do not have any common edges) is equal to the size of a minimal (S,T)-cut .*

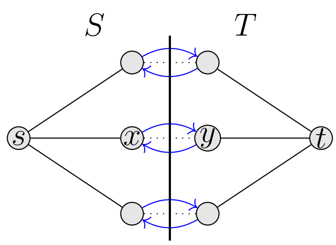
Proof. "Max flow-min cut theorem" (4.5) with $\forall e \in E \quad c(e) = 1$. □

Theorem 4.11 (Menger's theorem for edge-connectivity)**.** *Let G be a graph, $k \in \mathbb{N}$. G is k-edge-connected iff there are k edge-disjoint paths between any pair of vertices in G.*

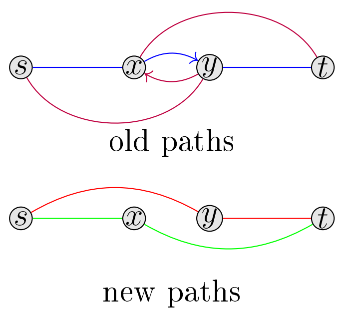
Proof.
" $\Longleftarrow$ " Removing $k-1$ edges will keep one edge intact, so the resulting graph is connected, hence G is k-edge connected.
" $\Longrightarrow$ " Let $s,t \in V, s \neq t$. Let G' be the graph obtained from G by replacing each edge by a dopple-edge (see graph(1)).

Claim: Any (s,t)-cut in G' has at least size k.
If (S,T) is an (s,t)-cut of size$\leq k-1$,then G has at most $k-1$ edges between S and T. Removing them disconnects the graph $\lightning$ G is k-edge connected. By Lemma (4.10), there are k-edge disjoint paths $p_1,...,p_k$ from s to t in G'.

Suppose $x,y \in V$ s.t. p_i uses $\{x,y\}$ and p_j uses $\{y,x\}$. Replace p_i and p_i by the path following p_i to x and continue via p_j and the path following p_j to y and continue via p_i. Repeat for each such edge. In each step, the sum of the lengths of the paths decreases so the process has to terminate. Then the underlying k-undirected paths in G are edge disjoint (see graph(2)).



graph(1)



graph(2)

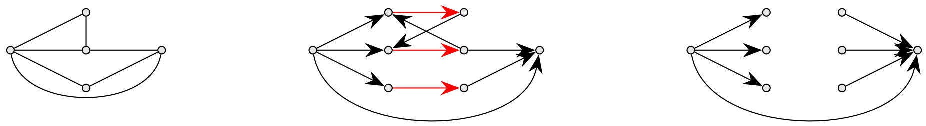
□

Theorem 4.12 (Menger's theorem for vertex-connectivity)**.** *Let G be a graph with at least $k+1$ vertices ($k \in \mathbb{N}$). Then G is k-vertex connected iff there are at least k internally disjoint paths (start and end are the only common vertices) between any pair of distinct vertices of G.*

Proof. " $\Longleftarrow$ " As before (Proof of (4.11)).
" $\Longrightarrow$ " Let $a,b \in V$ be distinct and $v_1,...,v_n$ be the remaining vertices of G. Consider the following directed Graph G' (see graph(1)):

- $V(G') = \{s,t,v_1',...,v_n',v_1'',...,v_n''\}$
- $E(G') = \{\{v_i',v_i''\} : i \in \{1,...,n\}\} \cup \{\{v_i'',v_j'\} : \{v_i,v_j\} \in E(G)\} \cup \{\{v_j'',v_i'\} : \{v_i,v_j\} \in E(G)\} \cup \{\{s,v_i\} : \{a,v_i\} \in E(G)\} \cup \{\{v_i'',t\} : \{b,v_i\} \in E(G)\} \cup \{\{s,t\}$ if $\{a,b\} \in E(G)\}$ where we call elements of $\{\{v_i',v_i''\} : i \in \{1,...,n\}\}$ red edges.

Let k_0 be the minimum size of an (s,t)-cut in G'. If $k_0 \geq k$, G' has k-edge disjoint paths from s to t. Since no red edge is contained in two of those, this corresponds to k internally disjoint paths from a to b in G: each path is of the form $(s, v_1', v_1'', ..., v_m', v_m'', t)$ and corresponds to $(a, v_i, ..., v_m, b)$ in G.
Now assume $k_0 < k$ and consider an (s,t)-cut (S,T) of G' of size k_0
Claim: We may assume that every edge in $\delta^+(S) \setminus \{s,t\}$ is red:
If an edge entering v_i' is in $\delta^+(S)$, move v_i' from T to S without changing the size of the cut. Similarly if an edge leaving v_i'' is in $\delta^+(S)$, move v_i'' to T. If $\{s,t\} \notin E(G)$, then removing the k_0 vertices of G that correspond to red edges in $\delta^+(S)$ disconnects G (no path from s to t in G' after removel $\implies$ none from a to b in G)$\lightning$.
If $\{s,t\} \in E(G)$, proceed as follows:
Let A be the set of vertices in G corresponding to red edges in $\delta^+(S)$ in G' ($|A| = k_0 - 1$). Let $\hat{G}$ be the graph obtained by removing A.
If $\hat{G}$ not connected $\lightning$ G is k-vertex connected.
Thus $\hat{G}$ connected. Since G has $\geq k+1$ vertices either a or b has an additional neighbour.
Wlog a does $\implies$ $\hat{G} - a$ is not connected $\lightning$ G is k-vertex connected (see graph(2)).



G $\qquad$ G' $\qquad$ $\hat{G}$

$\square$

Definition 4.13. A graph is **bipartite** with parts A, B if $V(G) = A \cup B$ and every edge in G has one en in A and one in B.

Example 4.14.

- $K_{a,b}$
- $Q_d, d \geq 1$

Definition 4.15. A **matching** in a graph G is a set $M \subset E(G)$ s.t. no two edges in M are incident.
A matching M is **perfect** if every vertex in G is incident to an edge in M.

Example 4.16.

perfect

maximal
not perfect

matching
not maximal
not perfect

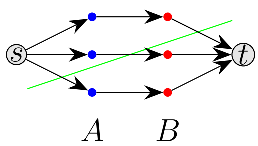
not matching

Theorem 4.17 (Hall's marriage theorem)**.** *Let $G = (V, E)$ be a bipartite graph with part A, B. Then G has a perfect matching iff $\forall P \subset A, |N_G(P)| \geq |P|$ and $\forall P \subset B, |N_G(P)| \geq |P|$.*

Proof. " $\implies$ " Assume (Wlog) $\exists P \subset A$ s.t. $|N_G(P)| < |P|$. Let M be a perfect matching. Then $\forall v \in P$ the other endpoint of the matching edge incident to v is in $N_G(v) \subset N_G(P)$, and these vertices are distinct $\implies |N_G(P)| \geq |P| \lightning$.
" $\impliedby$ " Note that the condition implies $|A| = |B|$, since $|B| \geq |N_G(A)| \geq |A|$ and vice versa. Define an auxiliary graph $G' = (V', E')$ with:

- $V' = V \cup \{s, t\}$
- $E' = \{\{s, a\} : a \in A\} \cup \{\{b, t\} : b \in B\} \cup \{\{a, b\} : a \in A, b \in B\}$
- $\forall e \in E : c(e) = 1$

Every matching M in G defines an integral (s, t)-flow ($f(e) \in \{0, 1\}$) in G with $\mathrm{val}(f) = |M|$ and vice versa.
Let (S, T) be an (s, t)-cut. Let $A_S = A \cap S, A_T = A \cap T, B_S = B \cap S, B_T = B \cap T$.
It follows $|\delta^+(S)| \geq |A_T| + |B_S| + |N_{A_S}(B_T)| \geq |N_{A_T}(B_T)| + |B_S| + |N_{A_S}(B_T)| = |N_A(B_T)| + |B_S| \overset{\text{condition}}{\geq} |B_T| + |B_S| = |B| = |A|$.
Thus by maxflow-mincut, $\max_{f \text{ flow}} \mathrm{val}(f) = \max_{f \text{ integral flow}} \mathrm{val}(f) \geq |B|$.
Therefore there exists a matching in G of size $\geq |B|$, i.e. a perfect matching.



$\square$

5 Coloring

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